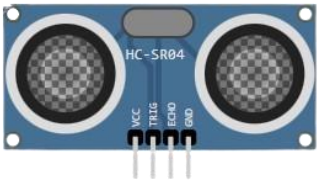
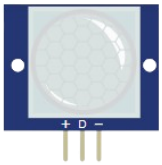
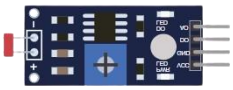
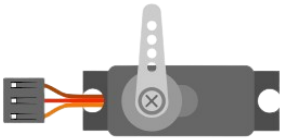
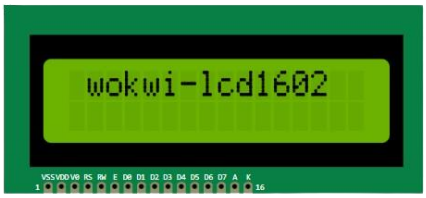
1 buah Mikrokontroler		
 ESP32		
3 Jenis Virtual Sensor		
 Ultrasonic Sensor (HC-SR04)	 PIR Motion Sensor	 LDR (Photoresistor)
2 Jenis Virtual Actuators		
 Servo Motor	 LED	
Tambahan		



LCD Display 16x2

Arsitektur IoT

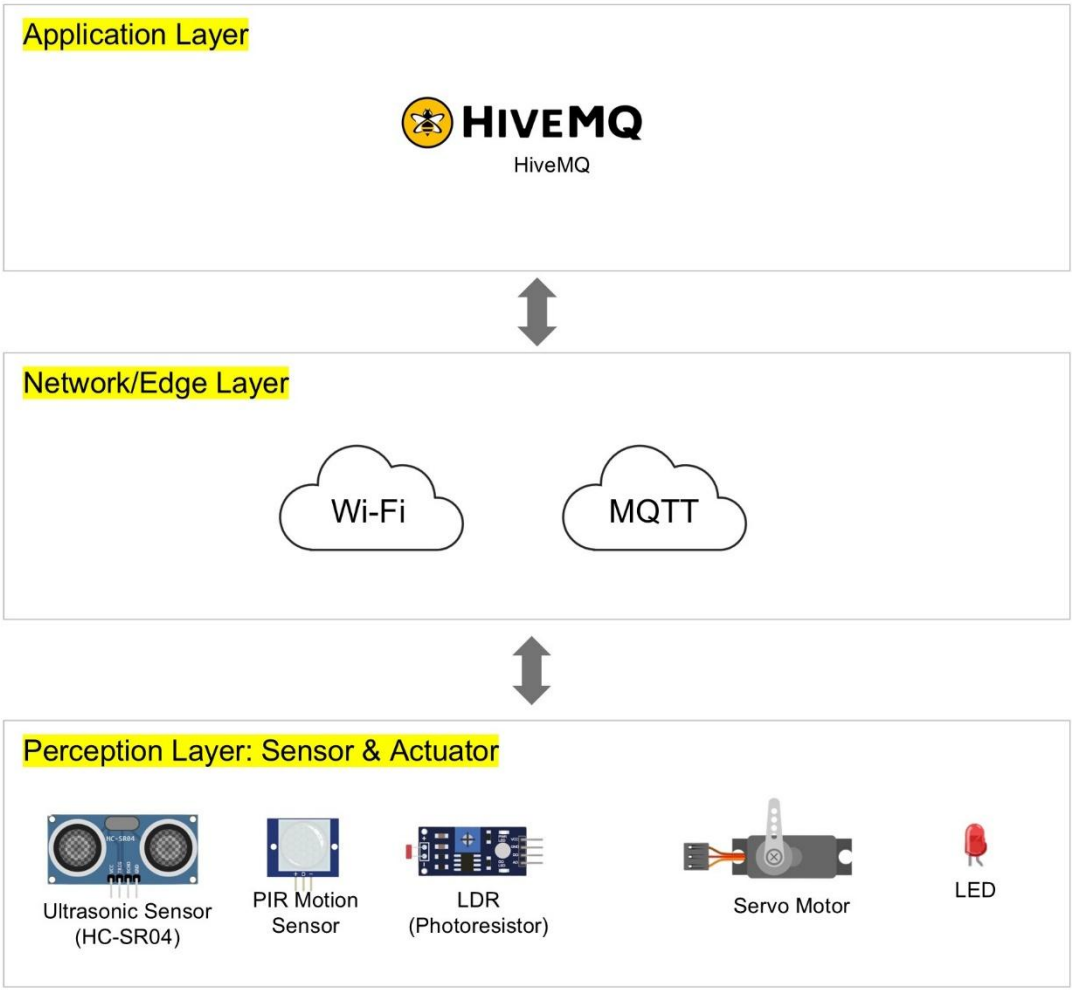
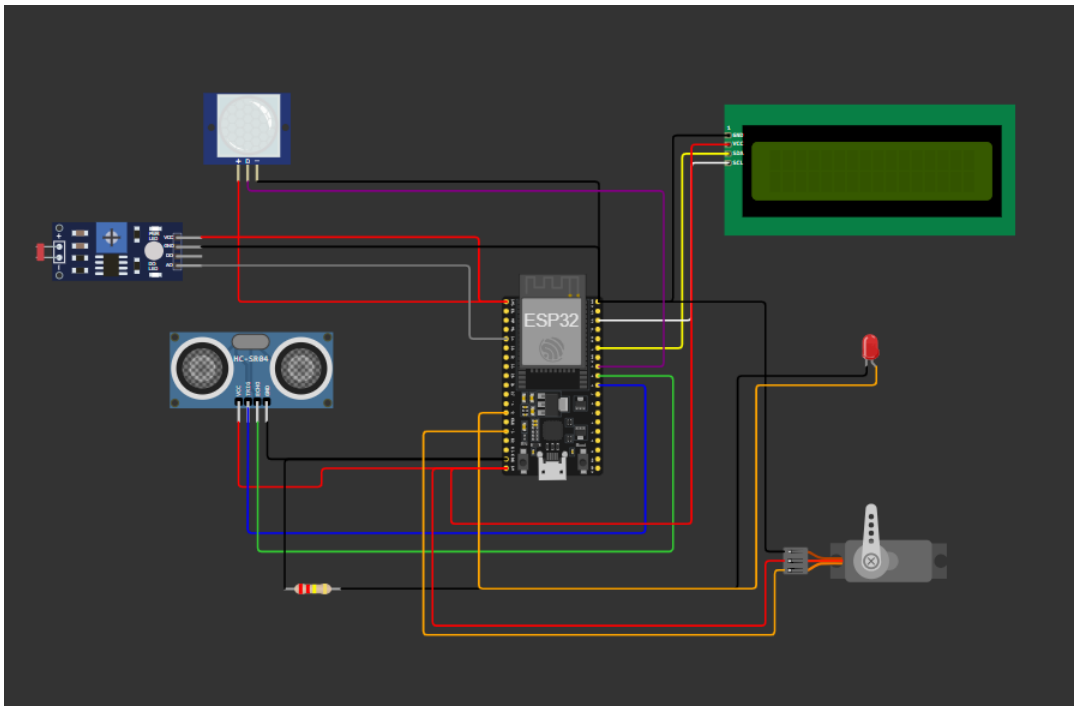


Diagram Wiring



Komponen	Nama Pin	Pin ESP32	Warna Kabel
HC-SR04	VCC	5V / VIN	Merah
	Trig	GPIO 5	Biru
	Echo	GPIO 18	Hijau
	GND	GND	Hitam
LCD 16x2 I2C	VCC	5V / VIN	Merah
	GND	GND	Hitam
	SDA	GPIO 21	Kuning
	SCL	GPIO 22	Putih
PIR Sensor	VCC (+)	3.3V	Merah
	OUT (D)	GPIO 19	Ungu
	GND (-)	GND	Hitam
Servo Motor	VCC (V+)	5V / VIN	Merah
	Signal (PWM)	GPIO 13	Oranye
	GND	GND	Hitam
LDR (Photoresistor)	VCC	3.3V	Merah
	AO (Analog)	GPIO 34	Abu-abu
	GND	GND	Hitam
LED	Anoda (+)	GPIO 12	Cokelat
	Katoda (-)	GND (via Resistor)	Hitam

FLOWCHART / WORKFLOW

Proses Inisialisasi

- ESP32 menyalakan LCD, menghubungkan ke WiFi Wokwi, dan menstabilkan koneksi ke MQTT Broker (HiveMQ).

Proses Buka/Tutup Palang Pintu Parkir:

- Sistem membaca data jarak dari Ultrasonic, nilai analog dari LDR, dan status logika dari PIR secara terus-menerus.
- Jika kendaraan mendekat (< 10cm), ESP32 langsung memerintahkan Servo ke 90° dan mengubah tampilan LCD menjadi "WELCOME".
- Sistem menunggu sensor PIR mendeteksi mobil telah lewat (HIGH) sebelum menutup kembali gerbang ke 0°.

Proses Menyalakan Cahaya

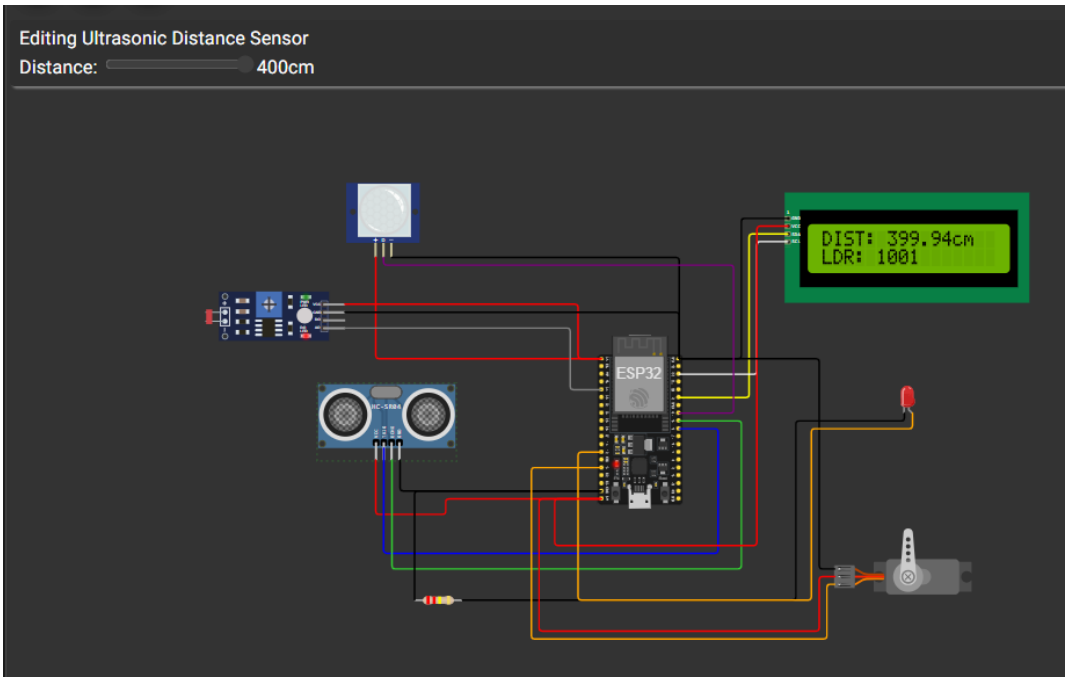
- Jika intensitas cahaya dari LDR turun di bawah ambang batas (gelap), ESP32 menyalakan LED secara otomatis tanpa menunggu instruksi cloud.

Proses Publishing

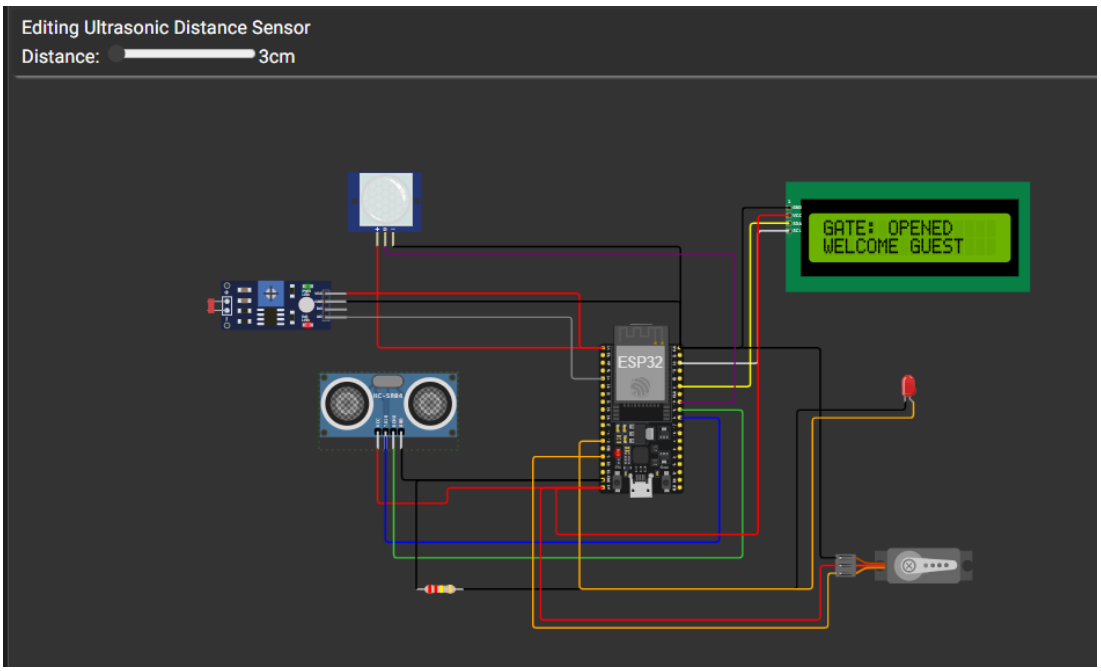
Semua data sensor dikonversi menjadi format JSON dan dikirim ke topik wokwi/smartparking/data di HiveMQ untuk pemantauan dashboard.

Visual (Screenshots)

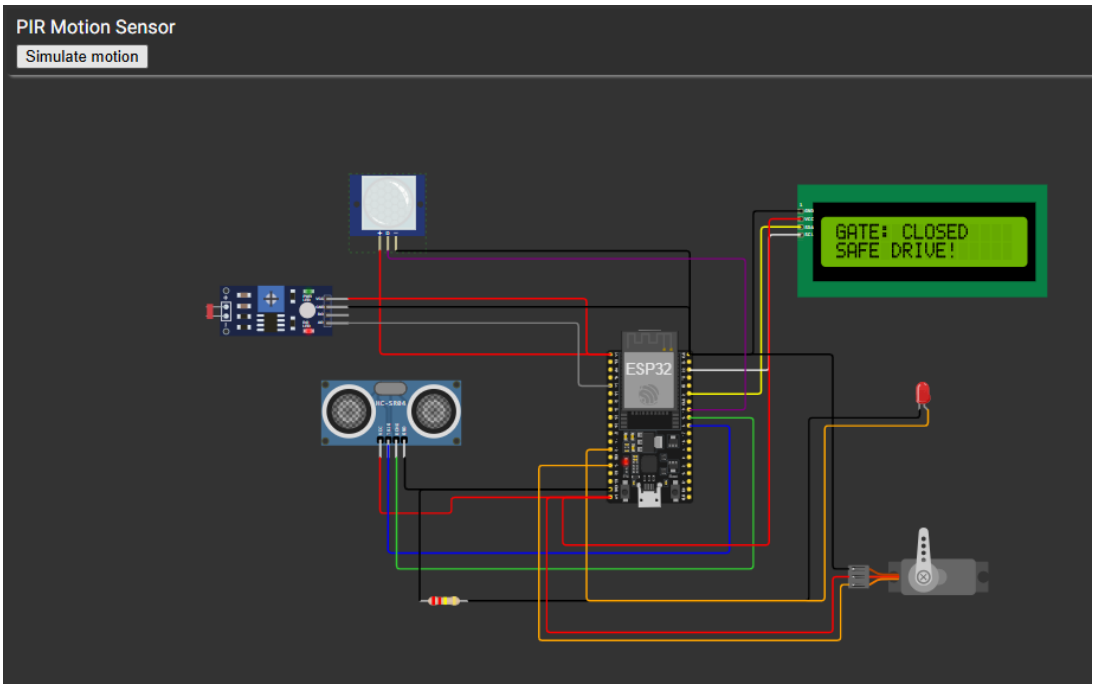
- Screenshot: Posisi Awal, Palang Pintu (Servo Motor) tertutup



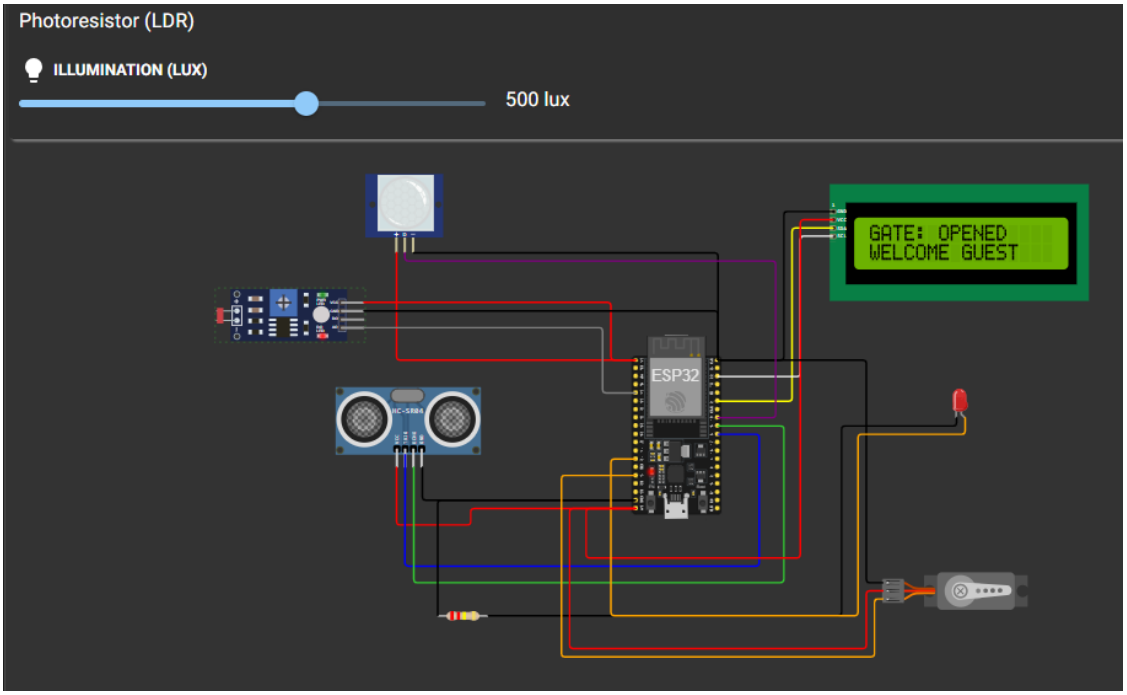
- Screenshot: Saat menggeser slider Ultrasonic ke < 10cm (LCD berubah jadi "Gate Opened").



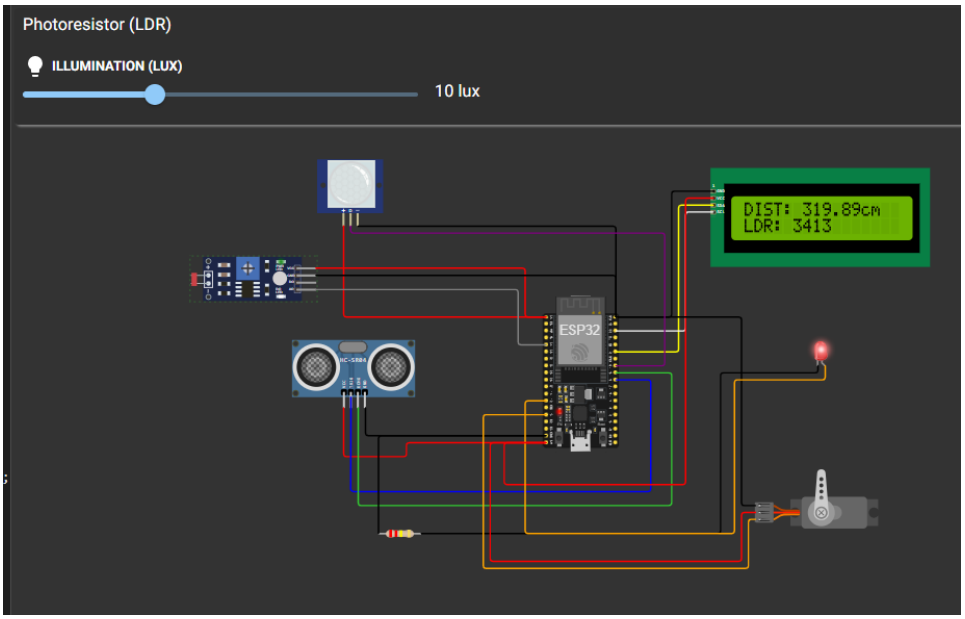
- Screenshot: Saat menggeser slider Ultrasonic ke > 10cm dan PIR Sensor Mendeteksi kendaraan suda lewat (LCD berubah jadi "Gate Closed").



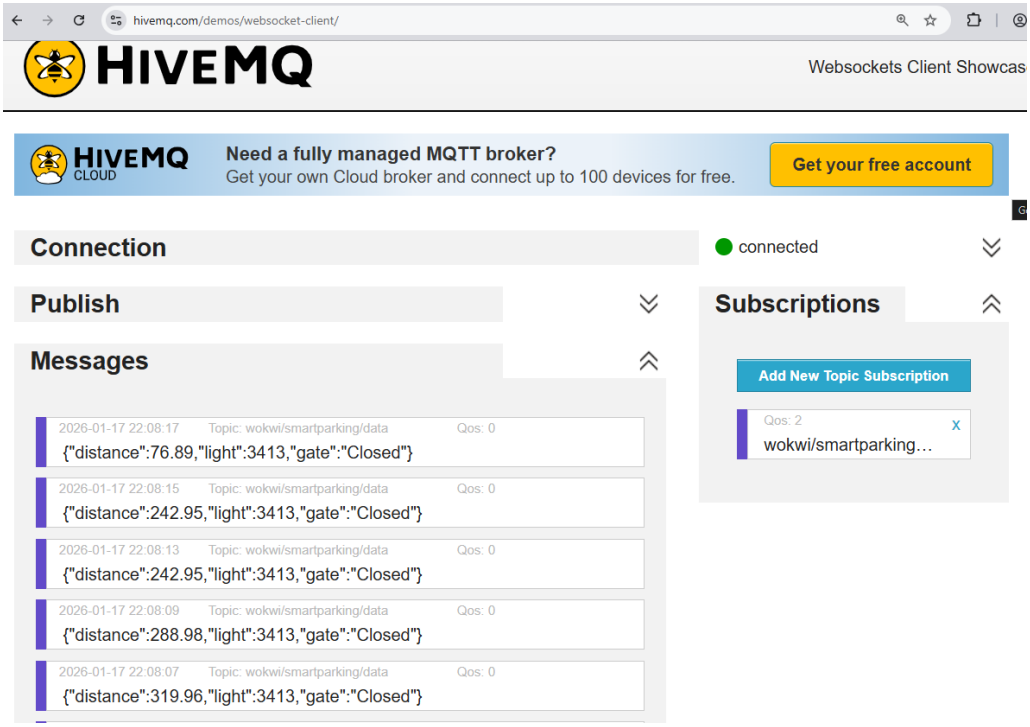
- Screenshot: LED tidak menyaka ketika slider LDR berada di posisi terang (analog > 2500).



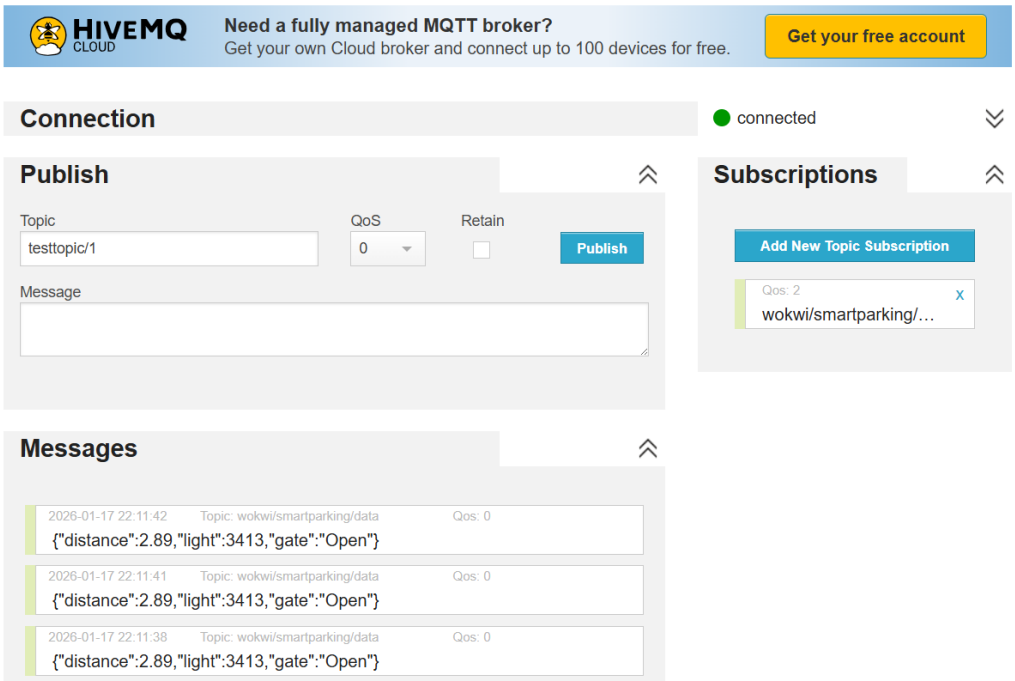
- Screenshot: Saat LED menyala ketika slider LDR berada di posisi gelap. (analog < 2500).



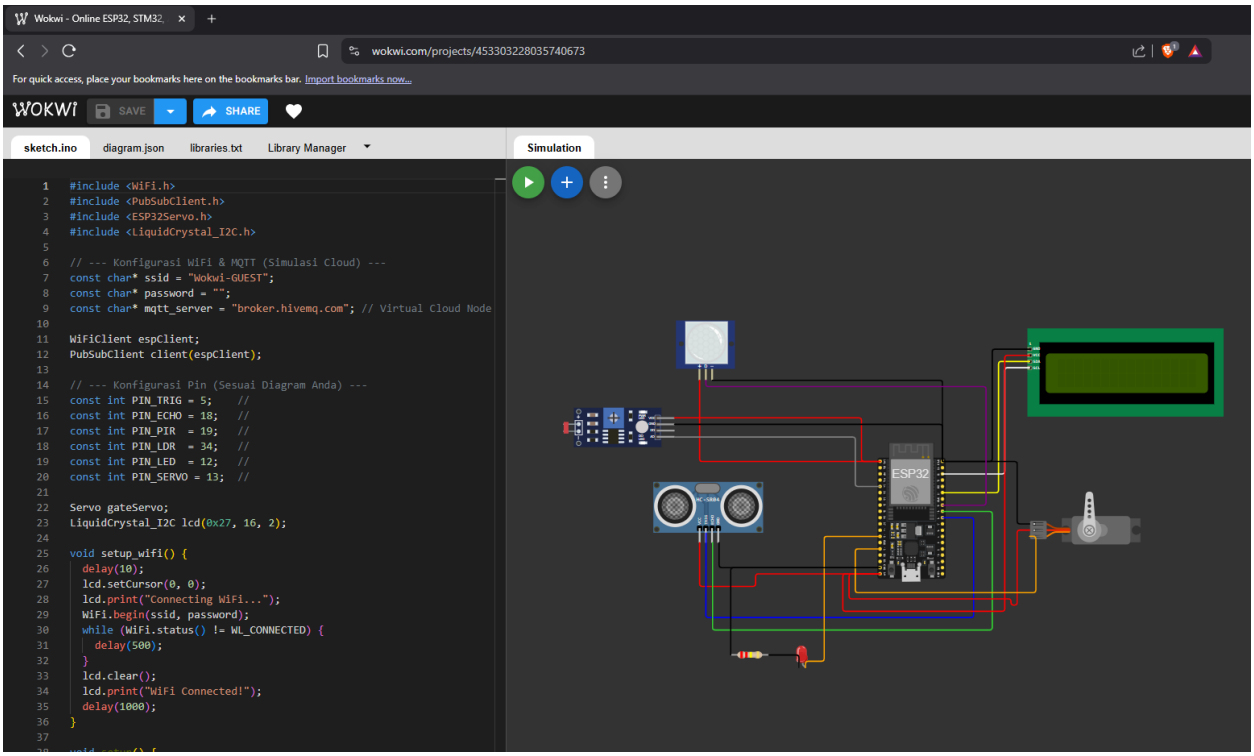
- Screenshot: Halaman HiveMQ Web Client yang menunjukkan “Gate Closed”



- Screenshot: Halaman HiveMQ Web Client yang menunjukkan “Gate Open”



LAMPIRAN – SIMULASI DI WORKWI



Situs Project dapat diakses di

https://wokwi.com/projects/453303228035740673

Situs HiveMQ dapat di akses di:

https://www.hivemq.com/demos/websocket-client/

Host : broker.hivemq.com
Port : 8884
Subscription : wokwi/smartparking/data